

PlutoDVB2

MQTT Reference Manual

Commands, payloads, status topics and practical examples

Firmware repository	https://github.com/F5OEO/plutosdr-fw
Bundled application	F5OEO/pluto-ori-ps commit c8a47400b86018a970c0df6e3825a9a99c7d4622
Firmware revision date	22 May 2025
Manual edition	1.1 - 5 September 2026

Official firmware and downloads

<https://github.com/F5OEO/plutosdr-fw>

This edition is matched to the exact pluto-ori-ps revision pinned by the firmware repository. Firmware behavior may change in later builds.

1. MQTT organization

PlutoDVB2 uses MQTT for control and telemetry. The configured callsign identifies the target Pluto.

```
Command: cmd/pluto/<CALL>/<topic>
Status: dt/pluto/<CALL>/<topic>
```

Replace **<CALL>** with the callsign stored in the firmware, for example **HB9IIU**.

Broker and ports

Topic	Payload / purpose
MQTT TCP	Port 1883 in the source configuration.
MQTT WebSocket	Port 9001 , path /wss, used by the included browser interface.
Subscription	Firmware processes subscribe to cmd/pluto/# and dt/pluto/#.

Requesting current values

For most normal commands, publish a question mark as the payload. The firmware answers on the corresponding dt/pluto/<CALL>/... topic.

```
mosquitto_pub -h 192.168.2.1 \
+ -t cmd/pluto/HB9IIU/tx/frequency -m '?'

mosquitto_sub -h 192.168.2.1 \
+ -t 'dt/pluto/HB9IIU/#' -v
```

Automatic command discovery

Publishing any payload to cmd/pluto/<CALL>/listcmd causes two lists to be published:

Topic	Payload / purpose
dt/pluto//listcmd_ctrl	Core RF control commands.
dt/pluto//listcmd_stream	Streaming and DVB modulator commands.

2. Core RF commands

Topic	Payload / purpose
rx/frequency	RX LO frequency in Hz. Numeric values may use the suffix K or M .
rx/modegain	RX gain-control mode: manual , slow_attack , hybrid or fast_attack .
rx/gain	Manual RX hardware gain in dB. Setting it also selects manual gain mode.
rx/sr	Requested RX sample rate in samples/s.
rx/format	0 = CS16; 1 = CS8; 2 = CS16 FFT.
rx/mute	1 powers down the RX LO; 0 enables it.
tx/frequency	TX LO frequency in Hz. Numeric values may use the suffix K or M .
tx/gain	TX hardware gain in dB, normally a negative value.
tx/sr	Internal TX sample rate. DVB symbol rate should normally be set with tx/dvbs2/sr.
tx/mute	1 = muted / TX off; 0 = unmuted / TX active.
tx/nco	Digital NCO frequency offset in Hz. Positive and negative values are accepted.

Frequency examples

```
# The following payloads represent the same frequency
2405000000
2405M
```

Caution

The meaning of tx/mute is easy to reverse accidentally: **1 disables RF output**; **0 enables RF output**.

3. DVB transmitter commands

Topic	Payload / purpose
tx/stream/run	1 runs the TX processing thread; 0 stops it.
tx/stream/mode	pass , iq , test , dvbs2-ts , dvbs2-gse or dvbs .
tx/dvbs2/sr	DVB symbol rate in symbols/s, for example 333000 .
tx/dvbs2/fec	Code rate; values are listed below.
tx/dvbs2/constel	qpsk , 8psk , 16apsk or 32apsk .
tx/dvbs2/frame	short or long .
tx/dvbs2/pilots	1 = enabled; 0 = disabled.
tx/dvbs2/fecmode	fixed or variable . Invalid input reports bad .
tx/dvbs2/fecrange	Integer from 0 through 10 .
tx/dvbs2/gainvariable	Variable-gain control value, normally 0 or 1 .
tx/dvbs2/digitalgain	Digital constellation gain below 3.0 . The source notes that it works only with QPSK.
tx/dvbs2/firfilter	0 or 1 selects the FPGA FIR-filter configuration.
tx/dvbs2/sdt	Service description/name inserted into the SDT. This is effectively write-only.
tx/dvbs2/rxbbbframeip	Multicast address and port for received BBFRAME data, e.g. 230.0.0.2:1234 .

Supported FEC payloads

1/4 1/3 2/5 1/2 3/5 2/3
3/4 4/5 5/6 8/9 9/10

Compact forms are also accepted: for example 23 for 2/3, 34 for 3/4 and 910 for 9/10.

4. Transport-stream input

Topic	Payload / purpose
tx/dvbs2/tssourcemode	0 = UDP; 1 = local file; 2 = generated pattern.
tx/dvbs2/tssourceaddress	UDP multicast/unicast address and port, e.g. 230.0.0.10:10000 . Applied only when source mode is 0.
tx/dvbs2/tssourcefile	Full transport-stream filename inside the Pluto. Applied only when source mode is 1.
tx/dvbs2/tssourcefilebitrate	File playback bitrate. Applied only when source mode is 1.

UDP DVB-S2 example

```
mosquitto_pub -h 192.168.2.1 \  
+ -t cmd/pluto/HB9IIU/tx/dvbs2/tssourcemode -m 0  
mosquitto_pub -h 192.168.2.1 \  
+ -t cmd/pluto/HB9IIU/tx/dvbs2/tssourceaddress \  
+ -m '230.0.0.10:10000'  
mosquitto_pub -h 192.168.2.1 \  
+ -t cmd/pluto/HB9IIU/tx/stream/mode -m dvbs2-ts
```

5. RX streaming and spectrum

Topic	Payload / purpose
rx/stream/run	1 runs RX processing; 0 stops it.
rx/stream/udp_addr_port	UDP destination address and port.
rx/stream/burst	RX burst size as an integer.
rx/stream/mode	pass or webfft .
rx/stream/average	Requested FFT averaging value. See known source issue in Section 9.
rx/webfft/frequency	Spectrum display centre frequency in Hz.
rx/webfft/span	Spectrum display span in Hz.

6. System and network commands

Topic	Payload / purpose
cmd/pluto/call	Sets a new callsign with <code>fw_setenv</code> , then reboots. This topic does not contain the current callsign in its path.
system/reboot	Any received payload triggers a reboot.
system/longmynd	Payload on starts the bundled Longmynd process; another payload stops it.
ip/tunadress	Address assigned to the <code>gse0</code> interface.
ip/route	Arguments passed directly to the Linux <code>route</code> command.
ip/iptables	Arguments passed directly to the Linux <code>iptables</code> command.

Security warning

The `ip/route` and `ip/iptables` payloads are executed as system-command arguments. Use them only on a trusted MQTT network.

Longmynd controls

These topics are handled by the bundled Longmynd build and do not use the Pluto callsign prefix.

Topic	Payload / purpose
cmd/longmynd/frequency	Receiver frequency in kHz.
cmd/longmynd/sr	Symbol rate in ksymbols/s.
cmd/longmynd/swport	Tuner input / switch port.
cmd/longmynd/tsip	Transport-stream multicast destination address.

7. Status and telemetry

Normal state is published beneath `dt/pluto/<CALL>/`. The following topics are generated explicitly by the examined source.

Topic	Payload / purpose
system/version	PlutoDVB2 version and source revision string.
system/device	Hardware model string.
temperature_ad	AD936x temperature reading.
rx/rssi	RX RSSI.
rx/frequency	Current RX LO frequency.
rx/modegain	Current RX gain-control mode.
rx/gain	Current RX hardware gain.
rx/finalsr	Resulting RX sample rate.
rx/fpgadecim	FPGA RX decimation factor.
rx/ad9361decim	AD936x RX decimation factor.
rx/stream/underflow	RX stream underflow count.
tx/frequency	Current TX LO frequency.
tx/gain	Current TX hardware gain.
tx/mute	Current TX LO power-down state.
tx/finalsr	Resulting TX sample rate.
tx/fpgainterpol	FPGA TX interpolation factor.
tx/ad9363interpol	AD936x TX interpolation factor.
tx/dvbs2/queue	Current BBFRAME queue depth.
tx/dvbs2/ts/bitrate	Calculated DVB-S2 TS bitrate.
tx/dvbs2/ts/fecvariable	Current variable FEC for TS mode.
tx/dvbs2/ts/ccerror	Most recently detected TS continuity-counter error.
tx/dvbs2/gse/fecvariable	Current variable FEC for GSE mode.
tx/dvbs2/gse/efficiency	Calculated GSE payload efficiency percentage.
tx/dvbs/ts/bitrate	Calculated DVB-S TS bitrate.
tx/dvbs/ts/ccerror	Most recently detected DVB-S TS continuity-counter error.
listcmd_ctrl	Comma-separated core-control command list.
listcmd_stream	Comma-separated stream/DVB command list.

8. Practical command sequence

The following example configures a DVB-S2 QPSK transmission. Review the gain and frequency before unmuting.

```
CALL=HB9IIU
HOST=192.168.2.1
ROOT=cmd/pluto/$CALL

mosquitto_pub -h $HOST -t $ROOT/tx/mute -m 1
mosquitto_pub -h $HOST -t $ROOT/tx/frequency -m 2405000000
mosquitto_pub -h $HOST -t $ROOT/tx/gain -m -30
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/sr -m 333000
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/fec -m 2/3
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/constel -m qpsk
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/frame -m short
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/pilots -m 1
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/tssourcemode -m 0
mosquitto_pub -h $HOST -t $ROOT/tx/dvbs2/tssourceaddress -m '230.0.0.10:10000'
mosquitto_pub -h $HOST -t $ROOT/tx/stream/mode -m dvbs2-ts
mosquitto_pub -h $HOST -t $ROOT/tx/mute -m 0
```

Monitor all Pluto telemetry

```
mosquitto_sub -h 192.168.2.1 -t 'dt/pluto/#' -v
```

9. Source-code observations

Topic	Payload / purpose
rx/stream/output_type	Advertised in the stream command list, but its handler is commented out.
tx/format	Advertised in the control command list, but no active handler is present.
tx/nco	Implemented in the application revision bundled by this firmware. It is commented out only in a later, currently unbundled source revision.
rx/stream/average	The setter changes the averaging variable, but the reply publishes the burst-size variable. This appears to be a source-code defect.
tx/dvbs2/sdt	The command updates the SDT but does not return a stored value when queried with ?.
tx/gain	Both the RF-control and stream processes receive this command: one sets AD936x hardware gain and the other records the maximum gain used by variable-gain logic.

10. Sources

Complete PlutoDVB2 firmware repository:

<https://github.com/F5OEO/plutosdr-fw>

MQTT/DVB application revision pinned by that firmware:

F5OEO, *pluto-ori-ps*, commit c8a47400b86018a970c0df6e3825a9a99c7d4622, 22 May 2025.

<https://github.com/F5OEO/pluto-ori-ps/tree/c8a47400b86018a970c0df6e3825a9a99c7d4622>

End of manual